Elias K. Oakes

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Education

The University of Connecticut — Storrs, CT
Ph.D. in Physics — *2024 – May 2027 (Expected)*
M.S. in Physics — 2021 – 2024
Advisor: Prof. Christopher M. Faesi

The University of Chicago — Chicago, IL
B.S. in Astrophysics with Honors; **B.A.** in Physics — 2016 – 2020
Advisor: Prof. Wendy L. Freedman

Awards and Honors

University of Connecticut Three-Minute Thesis finalist ($100)	2026
NASA CT Space Grant Consortium Graduate Fellowship ($10,000)	2026
University of Connecticut Doctoral Dissertation Fellowship ($2,000)	2026
Sigma Xi Honors Society (invited member)	2026
NRAO Student Observing Support Program ($40,000)	2025 – 2026
University of Connecticut Conference Participation Award ($750)	2024
NASA CT Space Grant Consortium Graduate Fellowship ($8,000)	2022
Chambliss Astronomy Achievement Award (honorable mention)	2021
National Merit Scholar ($16,000)	2016 – 2020
University of Chicago Dean's Fund for Undergraduate Research ($1,350)	2019
University of Chicago Dean's List	2016 – 2018

Professional Experience

Graduate Research Assistant — 2021 – Present
University of Connecticut, Storrs, CT
Studying the structure of the ISM and its relation to star formation in nearby galaxies using high-resolution ALMA, JWST, and MUSE data in the PHANGS and sSCALES collaborations.

Charge Injection Systems Expert — 2020 – 2021
CERN, Meyrin, Switzerland
Upgraded, calibrated, and maintained the Hadronic Tile Calorimeter on the ATLAS experiment at the Large Hadron Collider, with a focus on the charge injection and cooling systems.

Undergraduate Research Assistant — 2017 – 2020
Kavli Institute for Cosmological Physics, Chicago, IL
Measured three new distances to the Fornax dwarf spheroidal galaxy, working with Prof. Wendy Freedman as part of the Carnegie-Chicago Hubble Program.

Research Intern — 2019
REU Program at the Max Planck Institute for Astronomy, Heidelberg, Germany
Developed methods to associate giant molecular clouds and HII regions in overlapping ALMA and VLT/MUSE data, working with Prof. Christopher Faesi in the PHANGS collaboration.

Publications

21 papers (4 as first or second author); 280+ citations; h-index = 8.
Full list with individual contributions provided separately; see also the SciX library.

Talks and Presentations

Invited talks (13)	*Forthcoming fall 2026 seminar tour:* Yale, Carnegie Observatories, NRAO, UVa, Maryland, Stony Brook, Harvard CfA, Johns Hopkins, UCSD, Kansas, Columbia, Michigan, Rutgers	2026
Contributed poster	Northeast Star and Planet Formation meeting (Boston, MA)	2026
Contributed poster	European Astronomical Society meeting (Lausanne, Switzerland)	2026
Contributed talk	Northeast Star and Planet Formation meeting (New Haven, CT)	2025
Contributed talk	Galactic Ecosystems under the Microscope (Munich, Germany)	2025
Contributed talk	Views on the multi-phase ISM in galaxies (Bologna, Italy)	2025
Invited talk	Ohio State University Galaxy Coffee (Columbus, OH)	2024
Contributed talk	UNAM IRyA PhyISM meeting (Morelia, Mexico)	2024
Invited talk	MPIA Galaxy Coffee (Heidelberg, Germany)	2023
Contributed talk	Northeast Star and Planet Formation meeting (Cambridge, MA)	2023
Contributed poster	NASA CTSG Exposition (Windsor Locks, CT)	2022
Contributed talk	From Stars to Galaxies II (Gothenburg, Sweden)	2022
Invited talk	UConn Physics Colloquium (Storrs, CT)	2022
Contributed poster	237th meeting of the AAS (remote due to COVID-19)	2021
Contributed poster	235th meeting of the AAS (Honolulu, HI)	2020
Invited talk	MPIA Galaxy Coffee (Heidelberg, Germany)	2019
Contributed posters (2)	SPIE Optics and Photonics (San Diego, CA)	2018

17+ additional public outreach talks via PHANGS YouTube, Skype a Scientist, and Astronomy on Tap.

Service, Teaching, and Outreach

Students Mentored

Lia Gilmore, undergraduate thesis project, University of Connecticut 2024 – 2026
Neal Krishna, undergraduate thesis project, University of Connecticut 2022 – 2023

UConn Science, Technology, and Astronomy Recruits (STARs)

Managing Co-Lead 2024 – Present
Graduate Co-Lead 2022 – 2024

Mentor a cohort of 15–20 undergraduates from historically excluded backgrounds in physics. Led a published BiteScis lesson plan for high schools and co-authored program paper (AJP, in review).

PHANGS Colloquium and YouTube Committees

Founding Organizer 2022 – Present

Facilitate colloquia and produce an external video series about PHANGS research.

Skype a Scientist

Speaker (15 public talks and outreach events to date) 2022 – Present

Graduate Employee & Postdoc Union — UAW Local 6950

Organizing Committee 2024 – Present
Grievance Committee 2024 – Present
Union Guide (executive board) 2023 – 2024
District Steward 2022 – 2023

Teaching Experience (UConn Physics Department)

Course Assistant (PHYS 1025Q, Introductory Astronomy) 2024
Teaching Assistant (PHYS 1502Q, Physics for Engineers II) 2022
Teaching Assistant (PHYS 1401Q, General Physics I) 2021 – 2022

UConn Astrophysics Seminar Committee

Organizing Member 2022 – 2024

UConn Astronomy on Tap

Public Talk: "Historical Astronomy Through the Ages" 2022
Organizing Member 2021 – 2022

Successful Proposals and Observing Experience

As PI:

ALMA, 2026 (Cycle 13), 2026.1.01384.S — 75 hr/7m + 123 hr/TP
"The survival and evolution of high-altitude molecular gas in a unique nuclear outflow"

ALMA, 2025 (Cycle 12), 2025.1.01153.S — 4 hr/12m
"Revealing the vertical structure of molecular gas in a nearby Milky Way analog" (followup)

ALMA, 2024 (Cycle 11), 2024.1.01508.S — 22 hr/12m + 45 hr/7m + 80 hr/TP
"Revealing the vertical structure of molecular gas in a nearby Milky Way analog"

SMA, 2022 (student proposal for interferometry school) — 1 hr
"First detection of molecular cloud-scale dust continuum emission in M33"

As Co-I:

ALMA, 2026 (Cycle 13), 2026.1.00086.S, PI: Koch	7 hr/12m
JWST, 2026 (Cycle 5), 10046, PI: Leroy	200 hr
JWST, 2026 (Cycle 5), 10675, PI: Bolatto	34 hr
NOEMA, 2026 (Summer), S26BL, PI: He	20 hr
ALMA, 2025 (Cycle 12), 2025.1.00576.L, PI: Leroy	107 hr/12m + 216 hr/7m + 360 hr/TP
ALMA, 2025 (Cycle 12), 2025.1.00874.S, PI: Rosolowsky	31 hr/12m
ALMA, 2025 (Cycle 12), 2025.1.01449.S, PI: Rosolowsky	21 hr/12m + 68 hr/7m + 115 hr/TP
HST, 2025 (Cycle 33), 18086, PI: McClain	12 orbits
NOEMA, 2025 (Winter), W25BV, PI: He	20 hr
SMA, 2025A, S027, PI: den Brok	60 hr
NOEMA, 2024 (Winter), W24CJ, PI: He	16 hr
SMA, 2024B, S057, PI: Koch	270 hr
JWST, 2023 (Cycle 2), 3707, PI: Leroy	153 hr
JWST, 2023 (Cycle 2), 2987, PI: Leroy	23 hr
SMA, 2023A, S025, PI: Koch	50 hr
ALMA, 2022 (Cycle 10), 2022.1.00403.S, PI: Koch	39 hr/12m
IRAM 30m, 2022 (Winter), 155-22, PI: Faesi	16 hr

Executed observations at the telescope in February 2023.

Schools and Technical Skills

ALMA Data Reduction Workshop, Harvard-Smithsonian CfA (Cambridge, MA)	2024
Unraveling Galaxy Evolution with JWST, IMPRS (Heidelberg, Germany)	2023
Summer School on the ISM of Nearby Galaxies (Banyuls-sur-Mer, France)	2023
Synthesis Imaging Workshop, NRAO (Charlottesville, VA)	2023
Interferometry School, SMA (remote due to COVID-19)	2022
30-meter School on Millimeter Astronomy, IRAM (Granada, Spain)	2021

Programming: Proficient in Python, C, Java, HTML, ROOT, Linux systems, Git.

Software: Proficient with CASA, IRAF, CARTA, DAOPHOT/ALLSTAR, astrodendro, SCOUSE.

Languages and Citizenship

English (native), German (C1), French (B1), Spanish (A2).

Citizen of the United States and Switzerland.

Publications

21 papers (⋆4 as first or second author); 280+ citations; h-index = 8. See also the SciX library.

21. "UConn STARs: a program to support undergraduate physics students and engage in meaningful high school outreach." Battersby, C., Aguirre Serrata, E., **Oakes, Elias K.**, Willson, L., and STARs, submitted to *AJP*, in review (manuscript link).

⋆20. "The Imprint of Star Formation Seen on Parsec-scale Molecular Gas Profiles in NGC 253." **Oakes, Elias K.**, and the PHANGS collaboration, submitted to *ApJL*, in review (manuscript link).

19. "Quantifying the decline in CO(2–1) intensity around PHANGS-HST star clusters: Constraints on molecular gas dispersal, stellar feedback, and cluster formation efficiency." He, H. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2026, submitted to *A&A* (arXiv:2609.31838).

18. "Variations in the 3.3 μm Polycyclic Aromatic Hydrocarbon Feature Across Nearby Galaxies Driven by Metallicity and Radiation Field Spectrum." Koziol, H. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2026, *ApJ*, accepted (arXiv:2608.05286).

17. "A steep mass transition for bar-driven ISM structuring revealed by PHANGS-JWST." Emsellem, E. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2026, *A&A*, **713**, A155.

16. "Mid-Infrared Colors Vary with Galactic Environment: Contrasting Star-Forming Disks, Young Centers, and Quiescent Star-Formation Deserts." Pathak, D. et al. (incl. **Oakes, Elias K.**) 2026, *ApJ*, **1008**, 35.

15. "Multiscale Gas Structure and Dynamics in an Extragalactic Central Molecular Zone." Wang, L. et al. (incl. **Oakes, Elias K.**) 2026, *ApJ*, **1003**, 231.

14. "The Hidden Life of Stars: Embedded Beginnings to AGB Endings in the PHANGS-JWST Sample. I. Catalog of Mid-IR Sources." Hassani, H. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2026, *ApJS*, **284**, 3.

13. "Resolved HII Regions in NGC 253: Ionized Gas Structure and Suggestions of a Universal Density–Surface Brightness Relation." McClain, R. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2026, *ApJ*, **998**, 166.

12. "PAH Marks the Spot: Digging for Buried Clusters in Nearby Star-forming Galaxies." Graham, G. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2025, *AJ*, **170**, 340.

⋆11. "The Hierarchical Dynamical State of Molecular Gas from 3 to 300 pc in NGC 253." **Oakes, Elias K.**, and the PHANGS collaboration 2025, *ApJ*, **993**, 193.

⋆10. "Constraining Resolved Extragalactic R_{21} Variation with Well-calibrated ALMA Observations." den Brok, J., **Oakes, Elias K.**, and the PHANGS collaboration 2025, *ApJ*, **988**, 162.

9. "The MUSE view of the Sculptor galaxy: Survey overview and the planetary nebulae luminosity function." Congiu, E. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2025, *A&A*, **700**, A125.

8. "Cloud-scale Gas Properties, Depletion Times, and Star Formation Efficiency per Freefall Time in PHANGS–ALMA." Leroy, A. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2025, *ApJ*, **985**, 14.

7. "Polycyclic Aromatic Hydrocarbon and CO(2–1) Emission at 50–150 pc Scales in 70 Nearby Galaxies." Chown, R. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2025, *ApJ*, **983**, 64.

6. "The Fraction of Dust Mass in the Form of Polycyclic Aromatic Hydrocarbons on 10–50 pc Scales in Nearby Galaxies." Sutter, J. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2024, *ApJ*, **971**, 178.

5. "PHANGS-JWST: Data-processing Pipeline and First Full Public Data Release." Williams, T. G. et al. (incl. **Oakes, Elias K.**) and the PHANGS collaboration 2024, *ApJS*, **273**, 13.

4. "The impact of HII regions on giant molecular cloud properties in nearby galaxies sampled by PHANGS ALMA and MUSE." Zakardjian, A., Pety, J., Herrera, C. N., Hughes, A., **Oakes, Elias K.**, and the PHANGS collaboration 2023, *A&A*, **678**, A171.

*3. "Distances to Local Group Galaxies via Population II, Stellar Distance Indicators II: The Fornax Dwarf Spheroidal." **Oakes, Elias K.**, Hoyt, T. J., Freedman, W. L., Madore, B. F., Tran, Q. H., Cerny, W., Beaton, R. L., and Seibert, M. 2022, *ApJ*, **929**, 116.

2. "Distances to Local Group Galaxies via Population II, Stellar Distance Indicators I: The Sculptor Dwarf Spheroidal." Tran, Q. H., Hoyt, T. J., Freedman, W. L., Madore, B. F., **Oakes, Elias K.**, Cerny, W., Hatt, D., and Beaton, R. L. 2022, *ApJ*, **935**, 34.

1. "Multi-Wavelength, Optical (VI) and Near-Infrared (JHK) Calibration of the Tip of the Red Giant Branch Method based on Milky Way Globular Clusters." Cerny, W., Freedman, W. L., Madore, B. F., Ashmead, F., Hoyt, T. J., **Oakes, Elias K.**, Tran, Q. H., and Moss, B. 2020, arXiv:2012.09701.